CPE 4750: Introduction to IoT System, Spring 2026

Mid-Term Research Report

Title: Team Project Proposal

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I. Project Name

Our official, formal name for the Internet of Things Project is an AWS-Based Smart Energy-Harvesting Floor Tile.

II. Project Statement

The problem we're trying to solve is capturing energy from places with high foot traffic. This creates a decentralized and sustainable energy solution by easing the burden (from increased demand) on the current power grids.

III. IoT Components

The system consists of sensors, embedded processing, wireless communication, and cloud integration. Piezoelectric sensors collect voltage data generated from footsteps, which is processed by a microcontroller and transmitted to the cloud (AWS). The data is then stored, analyzed, and visualized in real time through a dashboard.

IV. Expected Deliverable

The final prototype will demonstrate a functional smart floor tile capable of detecting footsteps using piezoelectric sensors, measuring the generated voltage, and transmitting this data wirelessly to a central system. The system will display real-time foot traffic and estimated energy generation on an AWS-based dashboard, illustrating the complete pipeline from physical sensing to cloud-based visualization.

V. Project Plan

- Week 1: Hardware + Signal Processing + Logic
    - Set up piezoelectric sensor
    - Read analog signal on microcontroller
    - Turn raw signal (analog) into meaningful data (digital)
- Week 2: Simple UI + Microcontroller talks to Raspberry Pi
    - Develop a simple cloud UI
    - Establish communication between microcontroller and Raspberry Pi (ZigBee most likely)
- Week 3: Interim Demo
    - **Showcase:** Step on tile --> System detects event --> UI displays step count + voltage
    - **More detail:** Step on tile --> Signal Conditioning (rectification) --> Microcontroller --> (*hard-wired possibly*) --> Tx to Raspberry Pi Gateway --> Simple UI
- Week 4: Communication Layer
    - Implement network connectivity between ESP32 and Raspberry Pi
- Week 5: Dashboard + System Integration
    - Refine dashboard and enhance visualization
        - Showcase: Step Count + Energy (numerically + graphically)
    - Integrate full system pipeline
- Week 6: Final Demo
    - Debug and hardware/software issues
    - Prepare for presentation